Lecture 1
Introduction

Comp 206 – Introduction to Software Systems – Fall 2026

Mathieu Bérubé-Vallières

McGill University

September 1, 2026

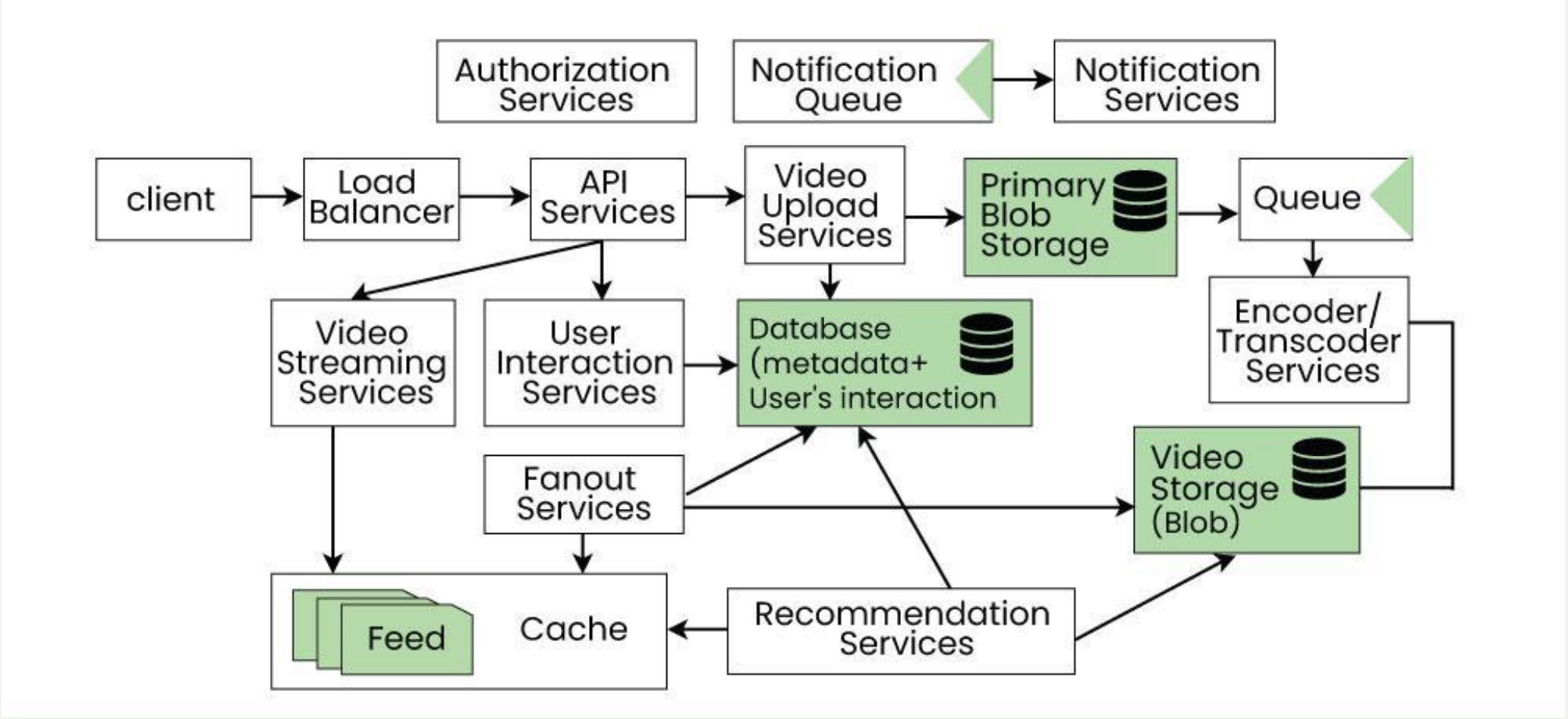
Software Systems

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What does it takes to build something like Instagram, Snapchat, or Tiktok?

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What does it takes to build something like Instagram, Snapchat, or Tiktok?



Software systems

A set of *interconnected*, *communicating* software artifacts, together with their specifications, configuration files, and documentation is called a **software system**.

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Systems development

- This course is an introduction to building large software
- You will learn essential tools used by systems developers
  - Linux, Command-line, Bash, git, C…COMP 206
- Sets the stage for several later courses in systems such as Operating Systems (COMP 310), Database Systems (COMP 421), Computer Networks (COMP 535), Distributed Systems (COMP 512), etc.

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Comp 206 specifics

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Mathieu Bérubé-Vallières



I am a faculty lecturer for SOCS

My research background is in quantum information theory! My work is on a topic called quantum pseudo-telepathy games.

Have a great semester :)



I am from Montreal



You may see me at the McGill Go Club!



This is my cat, Stella!

+1	+1	+1
+1	−1	−1
−1	+1	?

Ask me about quantum magic squares!

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TAs


Reza

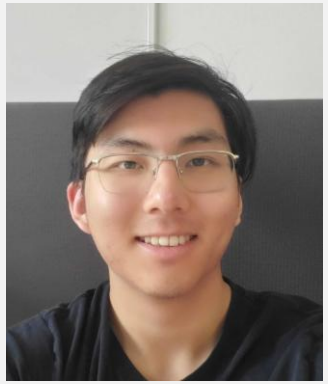

Siam


Eve


Maninder


Mohamad Ali


Thomas


Shiyan


Gurucharan


Abidur


Sara


Sabrina


Yongan (Michael) Yu

Lectures

- The class is taught in person
- Students are expected to be in Montreal and attend lectures
- Lecture recordings will be provided as an **additional study resource**
- They are done automatically and do not replace in-class attendance. Their availability and quality cannot be guaranteed.

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Course materials

- There is no required textbook
- The course schedule details which textbook sections correspond to each lecture. I will write them on the slides themselves too
  - (Note that some lectures contain additional material).
- Lectures will use slides and live coding demos. Slides will generally be released on MyCourses before classes.
- Please bring a computer to follow along!

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Prerequisite

- Comp 202 Foundations of Programming or Comp 250 Introduction to Computer Science

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Prerequisite

- Comp 202 Foundations of Programming or Comp 250 Introduction to Computer Science
- Quick test: Write a function that takes as input a list/array of integers as input and returns the index and value of the greatest element (in any programming language you know)

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Mimi

- All your work, notes, etc. in this course should be taken on the command-line on a server provided by the School of CS (SoCS) called *mimi*. This way you practice using the command-line!
- **Create your SoCS account** by visiting https://accounts.cs.mcgill.ca/ *while on McGill wifi* (or while connected to the McGill VPN) and *take note of the auto-generated password!* (You can later change it.)

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Assignments (20%)

- There are eight assignments, roughly weekly (2.5% each).
- Your homework assignments will be delivered via git and submitted via git (more on this soon).
  - They are to be completed on the Mimi server.
- Assignments are auto-graded and pass/fail.
  - To achieve the marks for an assignment, your submission must pass all of its test cases.
- Each homework assignment comes with a “soft” due date of the following week.
  - The “hard” due date for the assignments in module 1 is the module 1 midterm.
  - The “hard” due date for the assignments in module 2 is the module 2 midterm.

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In-class quizzes (10%)

- Lectures 2 to 8 (from Sep 3 to Sep 24) and lectures 12 to 20 (from Oct 8 to Nov 10) will contain 3 multiple choice questions on Vevox, i.e. 48 questions in total during the semester.
- Each **correct** answer is worth 5%, up to a maximum of 100% (i.e. getting 20 questions or more correct during the semester gives a grade of 100%, while getting 19 correct gives 95%, etc.).
- You must be in class to answer the questions. You will need to bring a phone or laptop to class to take the quiz.

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Exams

- Part 1 Exams – Fundamentals of Linux: 35%*
  - Class Test 1: 10%
    - 1h15, during class time. Covers Part 1.
  - Final Exam - Part 1: 25%
    - First half of the 3h final exam given during exam period. Same format as Class Test 1.
- Part 2 Exams – C Programming: 35%*
  - Class Test 2: 10%
    - 1h15, during class time. Covers Part 2.
  - Final Exam - Part 2: 25%
    - Second half of the 3h final exam given during exam period. Same format as Class Test 2.

*You must get 55% in both the “Part 1 Exams” and “Part 2 Exams” components to pass the course.

To make this more flexible, each of the two final sections corresponds to one of the class tests (same content covered, similar format with different questions), and we use the following weight shift system (see next slides for a visual explanation):

- If the score on a final exam section exceeds the corresponding classtest’s score, the full weight of that classtest shifts to that section of the final.
- If the score on a classtest exceeds the corresponding final exam section’s score, the full weight of that section of the final exam shifts to the classtest.

Weight shift example 1

Before weight shift:



Class test 1: 95%
Weight: 10%



Class test 2: 48%
Weight: 10%



Final section 1: 80%
Weight: 25%



Final section 2 : 86%
Weight: 25%



Before weight shift:
Part 1 Exams grade: 84,29%
Part 2 Exams grade: 75,14%
Total exams grade: 79,72%

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Weight shift example 1

After weight shift:



Class test 1: 95%
Weight: 35%



Class test 2: 48%
Weight: 0%



Final section 1: 80%
Weight: 0%



Final section 2 : 86%
Weight: 35%



After weight shift:
Part 1 Exams grade: 95%
Part 2 Exams grade: 86%
Total exams grade: 90,5%

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Weight shift example 2

Before weight shift:



Class test 1: 92%
Weight: 10%



Class test 2: 85%
Weight: 10%



Final section 1: 0%
Weight: 25%



Final section 2 : 0%
Weight: 25%



Before weight shift:
Part 1 Exams grade: 26,29%
Part 2 Exams grade: 24,29%
Total exams grade: 25,29%

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Weight shift example 2

After weight shift:



Class test 1: 92%
Weight: 35%



Class test 2: 85%
Weight: 35%



Final section 1: 0%
Weight: 0%



Final section 2 : 0%
Weight: 0%



After weight shift:
Part 1 Exams grade: 92%
Part 2 Exams grade: 85%
Total exams grade: 88.5%

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Weight shift example 2

After weight shift:



Class test 1: 92%
Weight: 35%



Class test 2: 85%
Weight: 35%



Final section 1: 0%
Weight: 0%



Final section 2 : 0%
Weight: 0%



After weight shift:
Part 1 Exams grade: 92%
Part 2 Exams grade: 85%
Total exams grade: 88.5%

ⓘ If you are satisfied with your class test grades, you do not have to go to the final!

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Class tests

- The class tests are tentatively planned to be held during class time on Tuesday October 6th and Thursday Nov 19th.
- Class Test 1 covers the first part of the course and Class Test 2 covers the second.
- You will have 1h15 to complete them.
- Location is TBD. Which room each student is assigned to will be announced closer to the class test date.
- If you have a conflict, notify me as soon as possible.
- This will be a closed book examination; A letter-sized (8.5x11in.) crib sheet will be permitted
  - Single or double-sided
  - Handwritten

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Final

- The final is administered by the Exam Office and will be held in person during the exam period (December 7 to December 22).
- The final examination consists of two evenly-weighted sections, one for each part of the course.
  - (See previous slides on weight-shifting with the class tests)
- You will have three hours to complete it.
- This will be a closed book examination; Two letter-sized (8.5x11in.) crib sheets will be permitted
  - Single or double-sided
  - Handwritten

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Course schedule

- The complete detailed course schedule with topics and tentative dates for quizzes, class tests, assignments, etc. is available on MyCourses->Content->Course Information
- The drop-in office hours (mine and TAs) will be available in MyCourses->Content->Weekly Schedule

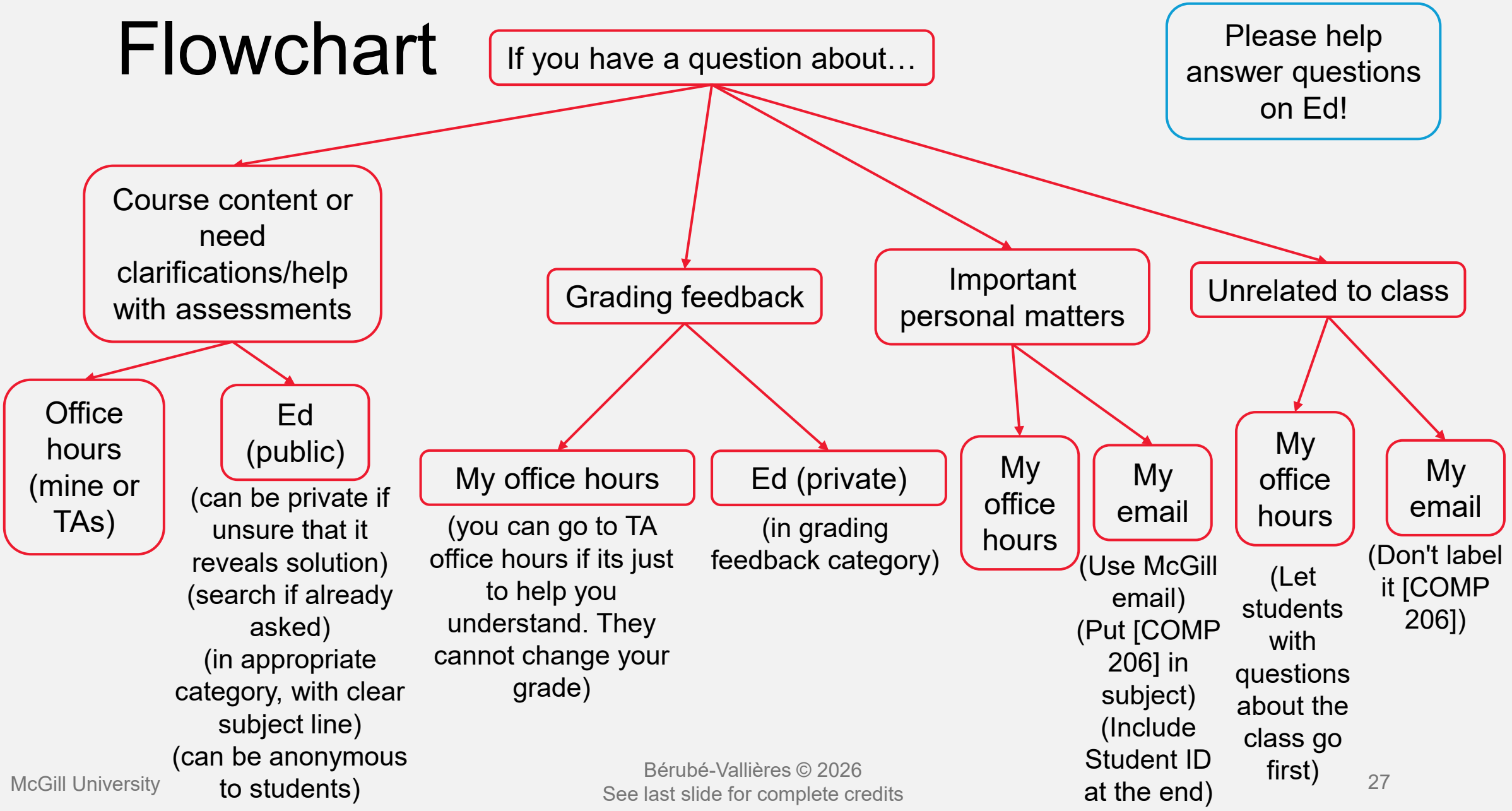
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Communication policies

- The main methods of communications are:
  - Prof drop-in office hours
    - 1.5 hours per week, Wednesdays 14:30-16:00 ENGMC 328
  - Prof appointment office hours
    - Click "Appointment" on MyCourses: I will give about 2.5 hours of appointment per week.
  - TA office hours TBD
  - The Ed discussion board
- My email can be used for:
  - Matters that are important and personal
  - Matters that are unrelated to the course (research, etc.)
- TA emails should never be used

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Flowchart



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Academic integrity

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Discussions in assignments

- I encourage you to discuss and brainstorm solutions together.
- However, individual copies should be your own work
  - It should not be copied or paraphrased
    - i.e. should not just edit someone else's work
  - You should be able to explain the reasoning behind your solutions
- In general, a good idea is to not leave discussions with other students with any written notes.
  - You should then wait a reasonable amount of time before writing your solution.
  - This ensures that the solution you write down is one that you can reconstruct by yourself from what you learned through the discussions.
  - You should not show copies of your work to any other student.

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Generative AI (ChatGPT)

You may use generative AI for:

- Learning and practicing the concepts in this course (e.g. help explain concepts, creating flashcards, planning your study timeline). As explained below, this does not include writing or double-checking solutions for any graded material.
- Spellchecking.

You may not use generative AI for:

- Completing any assessment, e.g. assignments, examinations (except spellchecking when applicable). In particular:
  - Copying or paraphrasing from generative AI and representing it as your own work is plagiarism
  - Double-checking the correctness of your solutions with generative AI is plagiarism

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Making class material available online

- You are allowed to:
  - Make your own class notes based on mine available online for non-commercial purposes (with attribution).
- You are not allowed to:
  - Share material written by me or the TAs online, including my lecture slides, assessment descriptions (assignments, exams), solutions, supplementary material, etc.
  - Share your solutions to assignments and exams online.
  - Share lecture recordings or any video of the class.

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Topics

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Two modules

Part 1: Fundamentals of Linux

Part 2: Programming in C

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Part 1: Fundamentals of Linux

At the end of this module, you will be able to:

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Part 1: Fundamentals of Linux

At the end of this module, you will be able to:

- Use the command-line on Linux to read, write, and manage files

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Part 1: Fundamentals of Linux

At the end of this module, you will be able to:

- Use the command-line on Linux to read, write, and manage files
- Compose atomic command-line programs into pipelines

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Part 1: Fundamentals of Linux

At the end of this module, you will be able to:

- Use the command-line on Linux to read, write, and manage files
- Compose atomic command-line programs into pipelines
- Write bash scripts to automate tasks relating to software development

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Part 1: Fundamentals of Linux

At the end of this module, you will be able to:

- Use the command-line on Linux to read, write, and manage files
- Compose atomic command-line programs into pipelines
- Write bash scripts to automate tasks relating to software development
- Use git to track modifications to bodies of source code

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Part 2: C Programming

At the end of this module, you will be able to:

- Write robust systems interactions in C
  - Memory management, garbage collection

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Part 2: C Programming

At the end of this module, you will be able to:

- Write robust systems interactions in C
- Use appropriate algorithms and data structures in C

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Part 2: C Programming

At the end of this module, you will be able to:

- Write robust systems interactions in C
- Use appropriate algorithms and data structures in C
- Write and build large programs with multiple files efficiently

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